

5. RL and RC First Order Response

Introduction to Electric Circuits Supplemental Instruction

Cooper McAllister

March 29 2026

1 First-Order Time Response

Circuits with DC sources and *only one* capacitor or inductor exhibit first-order response. If a change takes place at time $t = 0$ (such as a switch opening or closing), any current or voltage in the circuit, $m(t)$, can be found as:

$$m(t) = (m(0^+) - m(\infty))e^{\frac{-t}{\tau}} + m(\infty)$$

where

$m(0^+)$ is the quantity evaluated immediately after the change takes place,

$m(\infty)$ is the quantity evaluated a very long time after the change takes place, and

τ is the time constant:

For a circuit with a capacitor,

$$\tau = R_{th}C$$

For a circuit with an inductor,

$$\tau = \frac{L}{R_{th}}$$

R_{th} is the thevenin resistance of the circuit as seen from the capacitor or inductor.

A general procedure for finding the response of a first order circuit is as follows:

1. Draw the circuit before the change takes place (at $t = 0^-$) and replace capacitors and inductors with their steady-state equivalent
2. Find:
 - For a circuit with a capacitor: $V_C(0^-)$, the voltage across the capacitor. Then, $V_C(0^-) = V_C(0^+)$.
 - For a circuit with an inductor: $I_L(0^-)$, the current through the inductor. Then, $I_L(0^-) = I_L(0^+)$.

Draw the circuit after the change takes place and replace capacitors and inductors with their steady-state equivalent